

3(a)(i)	cell Y shows, a sieve plate / sieve pores ; ora idea that the sieve plates are at different heights in the stem / the section misses the sieve plate in cell X / AW ;	2
3(a)(ii)	any four from: 1 (protein in) companion cell surface membrane, moves / pumps, protons into, cell wall / apoplast ; 2 using ATP / by active transport ; 3 establishes a proton gradient ; 4 protons move into companion cell, by facilitated diffusion ; A described e.g. down the, electrochemical / concentration, gradient 5 protons cotransport sucrose / protons and sucrose pass through a cotransporter (protein), into companion cell ; 6 sucrose moves against its concentration gradient ; 7 diffusion of sucrose into phloem sieve tube, through plasmodesmata ; A all marking points from an annotated diagram	4

Question	Answer	Marks
3(b)(i)	this could be answered diagrammatically or in text, e.g. any two from: idea that oxygen has a small negative charge and hydrogen has a small positive charge ; idea that a hydrogen bond forms between the oxygen of one water molecule and the hydrogen of a neighbouring water molecule ; AVP ; e.g. oxygen is more electronegative than hydrogen	2
3(b)(ii)	any three from: 1 ref. to cohesion between water molecules ; 2 ref. to adhesion / hydrogen bonding, between water molecules and, cellulose / hydrophilic parts of lignin (in cell wall) ; A hydrophilic parts of cell wall 3 formation of continuous column / AW ; 4 (column) pulled up by, transpiration / evaporation / transpiration pull ;	3

Question	Answer	Marks
3(b)(iii)	any two from: ref. to plant leaves heat up, when light absorbed for photosynthesis / at high (ambient) temperatures ; (high latent heat of vaporisation leads to large cooling effect / removal of (large quantities of) heat energy, by evaporation ; reduces, protein / enzyme, denaturation ; reduces rate of water loss by, transpiration / evaporation, at high temperatures ; AVP ;	2

Question	Answer	Marks
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